

Andrea Amoretti

Curriculum vitae - self-hosted archival copy of the public University of Genoa CV

Fixed-term assistant professor
andrea.amoretti@unige.it

Education and training

2016

PhD in Theoretical Physics

AdS/CFT correspondence and its applications - Excellent
University of Genoa - Genoa - IT

2012

Master degree in Physics

3D Dynamics of 4D topological BF theory with boundary - 110/110 cum laude
University of Genoa - Genoa - IT

2010

Bachelor degree in Physics

Dynamics of non-holonomic constraints - 110/110 cum laude
University of Genoa - Genoa - IT

Academic experience

2018 - ongoing

Assistant professor (Ricercatore tempo determinato)

University of Genoa - Genoa - IT

2017 - 2018

Postdoctoral Research Associate

Universite Libre de Bruxelles - Bruxelles - BE

2016

Postdoctoral Research Associate

University of Wuerzburg - Wuerzburg - DE

2015 - 2016

Postdoctoral Research Associate

University of Cambridge - Cambridge - GB

Language skills

English - Proficient **French** - Independent **Italian** - Mother tongue

Source: University of Genoa public CV. The current UniGe staff directory lists Andrea Amoretti as Associate Professor; this archival PDF preserves the wording of the public CV snapshot.

Research interests

Strong field effects in BCS superconductors

The phenomenon of superconductivity is deeply connected with elementary-particle physics. Andrea studies the electrodynamics-superconductor analogy, including how strong electric fields modify superconductivity and whether phenomena analogous to the QED Schwinger effect can be realised in superconductors.

Hydrodynamical description of strongly coupled materials

Hydrodynamics is used as an effective field theory of long-lived, conserved or approximately conserved currents. The aim is to construct effective descriptions of strongly coupled electronic materials, including high-temperature superconductors, where ordinary particle-like quasiparticles may not provide the appropriate long-distance degrees of freedom.

Conformal field theories and perturbations around the critical point

Conformal field theories describe scale-invariant critical points of second-order phase transitions. Andrea combines conformal perturbation theory with numerical Monte Carlo methods to move away from criticality and study physical systems in the vicinity of a second-order transition.

Gauge/Gravity duality and strongly correlated systems

Gauge/Gravity duality relates strongly coupled quantum field theories to classical gravitational theories in a higher-dimensional spacetime. Andrea uses the correspondence to analyse difficult problems in strongly correlated condensed matter, including unconventional superconductors and fractional quantum Hall systems.

Research interests and grants

Topological Quantum Field Theories with boundary

Topological quantum field theories have global observables and provide effective descriptions of topological states of matter, including systems whose phases are not characterised by ordinary spontaneous symmetry breaking. Andrea has studied such theories in the presence of boundaries and their associated lower-dimensional dynamics.

Grants

2020 - ongoing

Curiosity Driven Project

EUR 63,000 - Principal Investigator

University of Genoa - IT

Public source

University of Genoa staff directory and public curriculum vitae: <https://rubrica.unige.it/personale/U0tHUI1g>

Archived for the STELLAR Lab website on 7 October 2026. The public CV source itself retains the older job-title wording; the live University of Genoa directory currently lists Andrea Amoretti as Associate Professor.